The Well Ordering Principle

August 12, 2025

Template for Well Ordering Proofs

There is a standard way to use Well Ordering to prove that some property, $P(n)$, holds for every nonnegative integer n. Here is a standard way to organize such a well ordering proof:

To prove that "$P(n)$ is true for all $n \in \mathbb{N}$" using the Well Ordering Principle:

1. Define the set C of counterexamples to P being true. Specifically, define

   $$C ::= \{n \in \mathbb{N} \mid \neg P(n) \text{ is true}\}.$$

   (The notation $\{n \mid Q(n)\}$ means "the set of all elements n for which $Q(n)$ is true." See Section 4.1.4.)

2. Assume for proof by contradiction that C is nonempty.

3. By the Well Ordering Principle, there will be a smallest element n in C.

4. Reach a contradiction somehow—often by showing that $P(n)$ is actually true or by showing that there is another member of C that is smaller than n. This is the open-ended part of the proof task.

5. Conclude that C must be empty, that is, no counterexamples exist.